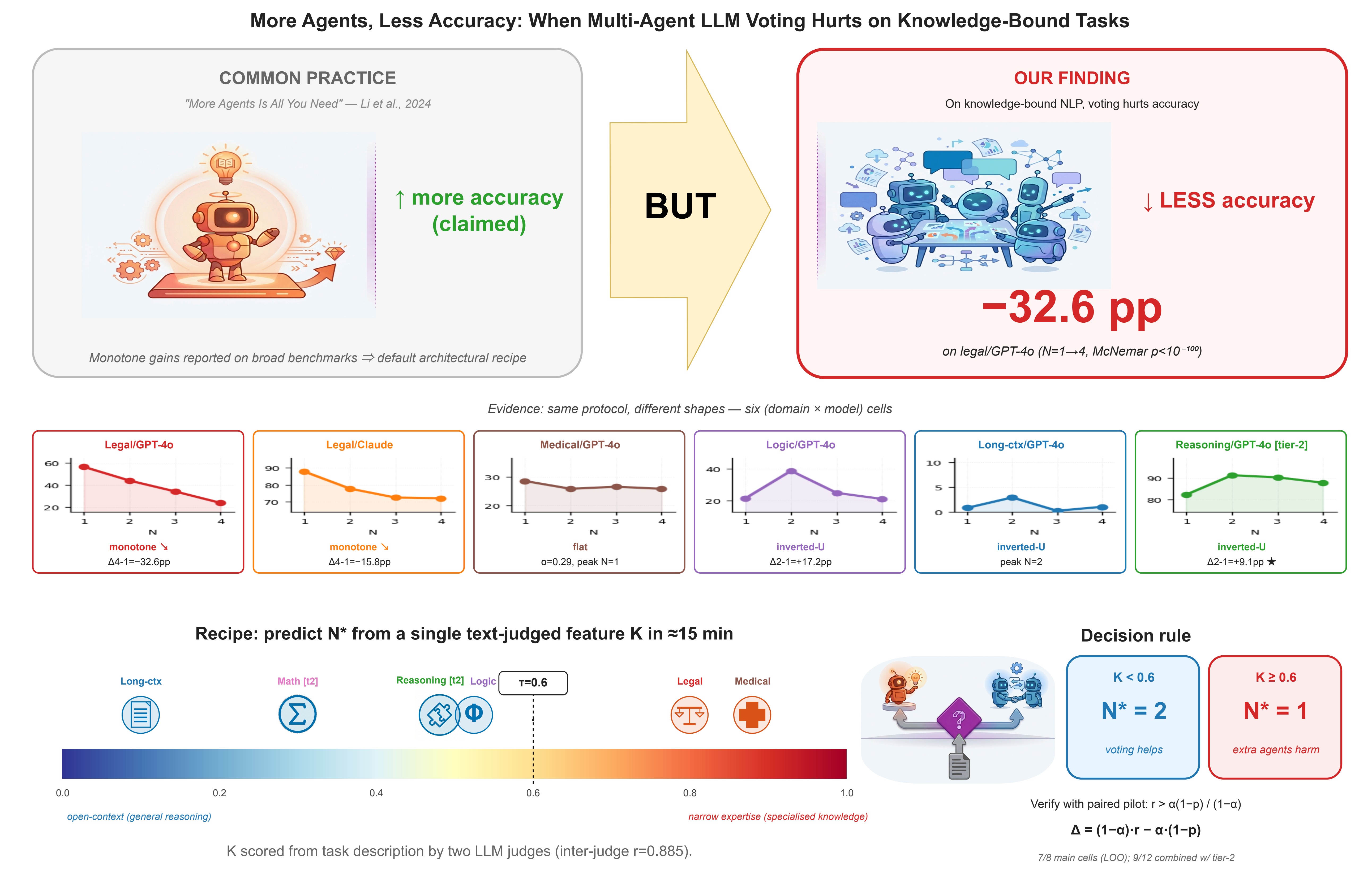
More Agents, Less Accuracy: When Multi-Agent LLM Voting Hurts on Knowledge-Bound Tasks
COMMON PRACTICE
"More Agents Is All You Need" — Li et al., 2024
↑ more accuracy (claimed)
Monotone gains reported on broad benchmarks ⇒ default architectural recipe
BUT
OUR FINDING
On knowledge-bound NLP, voting hurts accuracy
↓ LESS accuracy
−32.6 pp
on legal/GPT-4o (N=1→4, McNemar p<10⁻¹⁰⁰)
Evidence: same protocol, different shapes — six (domain × model) cells
Legal/GPT-4o
monotone ↘
Δ4-1=−32.6pp
Legal/Claude
monotone ↘
Δ4-1=−15.8pp
Medical/GPT-4o
flat
α=0.29, peak N=1
Logic/GPT-4o
inverted-U
Δ2-1=+17.2pp
Long-ctx/GPT-4o
inverted-U
peak N=2
Reasoning/GPT-4o [tier-2]
inverted-U
Δ2-1=+9.1pp ★
N
Recipe: predict N* from a single text-judged feature K in ≈15 min
Long-ctx
Math [t2]
Reasoning [t2]
Logic
τ=0.6
Legal
Medical
0.0
0.2
0.4
0.6
0.8
1.0
open-context (general reasoning)
narrow expertise (specialised knowledge)
K scored from task description by two LLM judges (inter-judge r=0.885).
Decision rule
K < 0.6
N* = 2
voting helps
K ≥ 0.6
N* = 1
extra agents harm
Verify with paired pilot: r > α(1−p) / (1−α)
Δ = (1−α)·r − α·(1−p)
7/8 main cells (LOO); 9/12 combined w/ tier-2